```
> restart:
  unprotect('D'):
   m := 4;
   twist := -7;
   d := 18;
  degre := d+1:

  TT := x^(d+1) + 1^(d+1) + 1*x^floor(d/2);

  Jetyx := expand(
      + (1/Rz^(m-0*2))*add(AA||j*x1^(m-0*3-j)*(y1)^j*Delta_yx^0, j=0..m
     -0*3)
      + (1/Rz^(m-1*2))*add(BB||j*x1^(m-1*3-j)*(y1)^j*Delta_yx^1, j=0..m
     -1*3)
      + (1/Rz^(m-2*2))*add(CC||j*x1^(m-2*3-j)*(y1)^j*Delta_yx^2, j=0..m
     -2*3)
      + (1/Rz^(m-3*2))*add(DD||j*x1^(m-3*3-j)*(y1)^j*Delta_yx^3, j=0..m
     -3*3)
   );

  dA := (m-0*2)*d - (m-0*3)*2 - 0*3 + twist;
  dB := (m-1*2)*d - (m-1*3)*2 - 1*3 + twist;
  dC := (m-2*2)*d - (m-2*3)*2 - 2*3 + twist;
  dD := (m-3*2)*d - (m-3*3)*2 - 3*3 + twist;

  Delta_yx := y1*x2 - y2*x1;
  Soly1 := -x1*Rx/Ry - z1*Rz/Ry;
  Soly2 := -x1^2*Rxx/Ry + 2*x1^2*Rxy*Rx/Ry^2 + 2*x1*Rxy*z1*Rz/Ry^2
     - 2*x1*z1*Rxz/Ry
           - Ryy*x1^2*Rx^2/Ry^3 - 2*Ryy*x1*Rx*z1*Rz/Ry^3 - Ryy*z1^2
     *Rz^2/Ry^3
           + 2*z1*Ryz*x1*Rx/Ry^2 + 2*z1^2*Ryz*Rz/Ry^2 - z1^2*Rzz/Ry
     - x2*Rx/Ry - z2*Rz/Ry;

  Jetzx := sort(mtaylor(expand(subs(x2=(Deltazx+x1*z2)/z1, expand(subs({y1
     =Soly1, y2=Soly2}, Jetyx)))), [x1, z1, Deltazx], 100), [x1, z1, Deltazx]);


  printlevel := 2:
  for k from 0 to m/3 do
      for j from 0 to m-3*k do
          EEEq[m-j-3*k, j, k] := factor(Rz^(m-j-3*k) * Ry^(m-2*k)
     * coeftayl(Jetzx, [x1, z1, Deltazx]=[0, 0, 0], [m-j-3*k, j, k]));
      od;
  od:
```

$$m := 4$$

$$twist := -7$$

$$d := 18$$

$$TT := x^{19} + x^{9} + 1$$

$$Jetyx := \frac{AA0\ x1^4}{Rz^4} + \frac{AA1\ x1^3\ y1}{Rz^4} + \frac{AA2\ x1^2\ y1^2}{Rz^4} + \frac{AA3\ x1\ y1^3}{Rz^4} + \frac{AA4\ y1^4}{Rz^4} + \frac{BB0\ x1\ Delta_yx}{Rz^2} + \frac{BB1\ y1\ Delta_yx}{Rz^2}$$

$$dA := 57$$

$$dB := 24$$

$$dC := -9$$

$$dD := -42$$

$$Delta_yx := -y2\ x1 + y1\ x2$$

$$Soly1 := -\frac{x1\ Rx}{Ry} - \frac{z1\ Rz}{Ry}$$

$$Soly2 := -\frac{x1^2\ Rxx}{Ry} + \frac{2\ x1^2\ Rxy\ Rx}{Ry^2} + \frac{2\ x1\ Rxy\ z1\ Rz}{Ry^2} - \frac{2\ x1\ z1\ Rxz}{Ry} - \frac{Ryy\ x1^2\ Rx^2}{Ry^3} - \frac{2\ Ryy\ x1\ Rx\ z1\ Rz}{Ry^3} - \frac{Ryy\ z1^2\ Rz^2}{Ry^3} + \frac{2\ z1\ Ryz\ x1\ Rx}{Ry^2} + \frac{2\ z1^2\ Ryz\ Rz}{Ry^2} - \frac{z1^2\ Rzz}{Ry} - \frac{x2\ Rx}{Ry} - \frac{z2\ Rz}{Ry}$$

$$Jetzx := \frac{1}{Rz^4\ Ry^4}\big((BB0\ Ryy\ Rx^2\ Rz^2\ Ry - 2\ BB0\ Rxy\ Rx\ Rz^2\ Ry^2 + BB0\ Rxx\ Rz^2\ Ry^3 - BB1\ Rx^3\ Ryy\ Rz^2 + 2\ BB1\ Rx^2\ Rxy\ Rz^2\ Ry - BB1\ Rx\ Rxx\ Rz^2\ Ry^2 + AA0\ Ry^4 - AA1\ Rx\ Ry^3 + AA2\ Rx^2\ Ry^2 - AA3\ Rx^3\ Ry + AA4\ Rx^4)\ x1^4\big) - \frac{1}{Rz^3\ Ry^4}\big((2\ BB0\ Ryz\ Rx\ Rz\ Ry^2 - 2\ BB0\ Ryy\ Rx\ Rz^2\ Ry + 2\ BB0\ Rxy\ Rz^2\ Ry^2 - 2\ BB0\ Rxz\ Rz\ Ry^3 - 2\ BB1\ Rx^2\ Ryz\ Rz\ Ry + 3\ BB1\ Rx^2\ Ryy\ Rz^2 - 4\ BB1\ Rx\ Rxy\ Rz^2\ Ry + 2\ BB1\ Rx\ Rxz\ Rz\ Ry^2 + BB1\ Rxx\ Rz^2\ Ry^2 + AA1\ Ry^3 - 2\ AA2\ Rx\ Ry^2 + 3\ AA3\ Rx^2\ Ry - 4\ AA4\ Rx^3)\ x1^3\ z1\big) + \frac{1}{Rz^2\ Ry^4}\big((BB0\ Rzz\ Ry^3 - 2\ BB0\ Ryz\ Rz\ Ry^2 + BB0\ Ryy\ Rz^2\ Ry - BB1\ Rx\ Rzz\ Ry^2 + 4\ BB1\ Rx\ Ryz\ Rz\ Ry - 3\ BB1\ Rx\ Ryy\ Rz^2 + 2\ BB1\ Rxy\ Rz^2\ Ry - 2\ BB1\ Rxz\ Rz\ Ry^2 + AA2\ Ry^2 - 3\ AA3\ Rx\ Ry + 6\ AA4\ Rx^2)\ x1^2\ z1^2\big) - \frac{(BB1\ Rzz\ Ry^2 - 2\ BB1\ Ryz\ Rz\ Ry + Rz^2\ BB1\ Ryy + AA3\ Ry - 4\ AA4\ Rx)\ x1\ z1^3}{Rz\ Ry^4} + \frac{AA4\ z1^4}{Ry^4} - \frac{(BB0\ Ry - BB1\ Rx)\ x1\ Deltazx}{Rz\ Ry^2} + \frac{BB1\ z1\ Deltazx}{Ry^2} \tag{1}$$

```
> ## EQUATION RR EXPLICITE
  RR := z^(d+1) + y^(d+1) + TT;
  Rx := diff(RR, x): Ry := diff(RR, y): Rz := diff(RR, z);
```

```
Rxx := diff(RR, x, x): Rxy := diff(RR, x, y): Rxz := diff(RR, x, z);
Ryy := diff(RR, y, y): Ryz := diff(RR, y, z): Rzz := diff(RR, z, z);

printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEEq[m-j-3*k, j, k] := mtaylor(EEEq[m-j-3*k, j, k], z, (m-j-3
    *k)*d);
    od;
od:

for j1 from 0 to 1+m*d+d do
    ny[j1] := (-z^(d+1)-TT)^(floor(j1/(d+1))) * y^(j1-(d+1)*floor(j1
    /(d+1)));
od:

for l from 0 to m-0*3 do
    AA||l := add(A||l[j]*y^j, j=0..min(d, dA));
od:
for l from 0 to m-1*3 do
    BB||l := add(B||l[j]*y^j, j=0..min(d, dB));
od:

for l from 0 to m-2*3 do
    CC||l := add(C||l[j]*y^j, j=0..min(d, dC));
od:
for l from 0 to m-3*3 do
    DD||l := add(D||l[j]*y^j, j=0..min(d, dD));
od:

1+m*d+d;

printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        degree(EEEq[m-j-3*k, j, k], y);
        EEEq[m-j-3*k, j, k] := mtaylor(EEEq[m-j-3*k, j, k], y, 1+m*d
    +d);
        dy[m-j-3*k, j, k] := degree(EEEq[m-j-3*k, j, k], y);
    od;
od:

printlevel := 3:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        for j1 from 0 to dy[m-j-3*k, j, k] do
            Cy[m-j-3*k, j, k, j1] := coeff(EEEq[m-j-3*k, j, k], y, j1);
        od;
    od;
```

```
od:

printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k, j, k] := add(Cy[m-j-3*k, j, k, j1]*ny[j1], j1=0
    ..dy[m-j-3*k, j, k]);
        degree(EEq[m-j-3*k, j, k], y);
    od;
od:

printlevel := 2:
for l from 0 to m-0*3 do
    for j from 0 to min(d, dA) do
        A||l[j] := add(A||l[j, k]*z^k, k=0..dA-j);
    od;
od:

printlevel := 2:
for l from 0 to m-1*3 do
    for j from 0 to min(d, dB) do
        B||l[j] := add(B||l[j, k]*z^k, k=0..dB-j);
    od;
od:

printlevel := 2:
for l from 0 to m-2*3 do
    for j from 0 to min(d, dC) do
        C||l[j] := add(C||l[j, k]*z^k, k=0..dC-j);
    od;
od:

printlevel := 2:
for l from 0 to m-3*3 do
    for j from 0 to min(d, dD) do
        D||l[j] := add(D||l[j, k]*z^k, k=0..dD-j);
    od;
od:

printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k, j, k] := mtaylor(EEq[m-j-3*k, j, k], z, (m-j-3*k)
    *d);
        degree(EEq[m-j-3*k, j, k], z);
    od;
od:

printlevel := 3:
```

```
for l from 0 to m-0*3 do
    for j from 0 to min(d, dA) do
        for k from 0 to dA-j do
            A||l[j, k] := add(A||l[i, j, k]*x^i, i=0..dA-j-k);
        od;
    od;
od:

printlevel := 3:
for l from 0 to m-1*3 do
    for j from 0 to min(d, dB) do
        for k from 0 to dB-j do
            B||l[j, k] := add(B||l[i, j, k]*x^i, i=0..dB-j-k);
        od;
    od;
od:

printlevel := 3:
for l from 0 to m-2*3 do
    for j from 0 to min(d, dC) do
        for k from 0 to dC-j do
            C||l[j, k] := add(C||l[i, j, k]*x^i, i=0..dC-j-k);
        od;
    od;
od:

printlevel := 3:
for l from 0 to m-3*3 do
    for j from 0 to min(d, dD) do
        for k from 0 to dD-j do
            D||l[j, k] := add(D||l[i, j, k]*x^i, i=0..dD-j-k);
        od;
    od;
od:

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;

printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        Eq[m-j-3*k, j, k] := mtaylor(EEq[m-j-3*k, j, k], z, (m-j-3*k)
    *d);
        degree(Eq[m-j-3*k, j, k], z);
    od;
od:
```

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        degree(Eq[m-j-3*k, j, k], y);
    od;
od;
```

$$RR := x^{19} + y^{19} + z^{19} + x^{9} + 1$$

$$Rz := 19\ z^{18}$$

$$Rxz := 0$$

$$Rzz := 342\ z^{17}$$

$$\mathbf{91}$$

$$l := 0$$

$$\mathbf{23560}$$

$$l := 1$$

$$\mathbf{23560}$$

$$l := 2$$

$$\mathbf{23560}$$

$$l := 3$$

$$\mathbf{23560}$$

$$l := 4$$

$$\mathbf{23560}$$

$$l := 5$$

$$l := 0$$

$$\mathbf{2869}$$

$$l := 1$$

$$\mathbf{2869}$$

$$l := 2$$

$$l := 0$$

$$l := 0$$

$$k := 0$$

$$\mathbf{18}$$

$$\mathbf{18}$$

$$\mathbf{18}$$

$$\mathbf{18}$$

$$-\infty$$

$$k := 1$$

$$\mathbf{18}$$

$$-\infty$$

$$k := 2 \qquad \mathbf{(2)}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(1*d)
  Systemez1d := { }:
  for k from 0 to (m-1)/3 do
      Coeffs_z1d[1, m-1-3*k, k] := {coeffs(expand(mtaylor(Eq[1, m-1-3*k,
      k], z, 1*d)), [x, y, z])};
      Systemez1d := Systemez1d union Coeffs_z1d[1, m-1-3*k, k];
  od:
  Systeme_z1d := Systemez1d:
  Variables_z1d := indets(Systeme_z1d):
  Solutions_z1d := solve(Systeme_z1d, Variables_z1d):
  assign(Solutions_z1d):

  for l from 0 to m-0*3 do nops(expand(AA||l)) od;
  for l from 0 to m-1*3 do nops(expand(BB||l)) od;
  for l from 0 to m-2*3 do nops(expand(CC||l)) od;
  for l from 0 to m-3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$23560$$
$$l := 1$$
$$23560$$
$$l := 2$$
$$23560$$
$$l := 3$$
$$79570$$
$$l := 4$$
$$129427$$
$$l := 5$$
$$l := 0$$
$$284$$
$$l := 1$$
$$322$$
$$l := 2$$
$$l := 0$$
$$l := 0 \tag{3}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(2*d)
  Systemez2d := { }:
  for k from 0 to (m-2)/3 do
      Coeffs_z2d[2, m-2-3*k, k] := {coeffs(expand(mtaylor(Eq[2, m-2-3*k,
      k], z, 2*d)), [x, y, z])};
      Systemez2d := Systemez2d union Coeffs_z2d[2, m-2-3*k, k];
  od:
  Systeme_z2d := Systemez2d:
```

```
Variables_z2d := indets(Systeme_z2d):
Solutions_z2d := solve(Systeme_z2d, Variables_z2d):
assign(Solutions_z2d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$23560$$
$$l := 1$$
$$23560$$
$$l := 2$$
$$26302$$
$$l := 3$$
$$35336$$
$$l := 4$$
$$41613$$
$$l := 5$$
$$l := 0$$
$$256$$
$$l := 1$$
$$298$$
$$l := 2$$
$$l := 0$$
$$l := 0 \tag{4}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(3*d)
Systemez3d := { }:
for k from 0 to (m-3)/3 do
    Coeffs_z3d[3, m-3-3*k, k] := {coeffs(expand(mtaylor(Eq[3, m-3-3*k,
    k], z, 3*d)), [x, y, z])};
    Systemez3d := Systemez3d union Coeffs_z3d[3, m-3-3*k, k];
od:
Systeme_z3d := Systemez3d:
Variables_z3d := indets(Systeme_z3d):
Solutions_z3d := solve(Systeme_z3d, Variables_z3d):
assign(Solutions_z3d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$23560$$
$$l := 1$$
$$513$$
$$l := 2$$
$$714$$
$$l := 3$$
$$689$$
$$l := 4$$
$$488$$
$$l := 5$$
$$l := 0$$
$$201$$
$$l := 1$$
$$218$$
$$l := 2$$
$$l := 0$$
$$l := 0 \tag{5}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(4*d)
  Systemez4d := { }:
  for k from 0 to (m-4)/3 do
      Coeffs_z4d[4, m-4-3*k, k] := {coeffs(expand(mtaylor(Eq[4, m-4-3*k,
      k], z, 4*d)), [x, y, z])};
      Systemez4d := Systemez4d union Coeffs_z4d[4, m-4-3*k, k];
  od:
  Systeme_z4d := Systemez4d:
  Variables_z4d := indets(Systeme_z4d):
  Solutions_z4d := solve(Systeme_z4d, Variables_z4d):
  assign(Solutions_z4d):

  for l from 0 to m-0*3 do nops(expand(AA||l)) od;
  for l from 0 to m-1*3 do nops(expand(BB||l)) od;
  for l from 0 to m-2*3 do nops(expand(CC||l)) od;
  for l from 0 to m-3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$

$$\mathbf{1}$$

$$l := 4$$

$$\mathbf{1}$$

$$l := 5$$

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$l := 0$$

$$l := 0 \tag{6}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(5*d)
  Systemez5d := { }:
  for k from 0 to (m-5)/3 do
      Coeffs_z5d[5, m-5-3*k, k] := {coeffs(expand(mtaylor(Eq[5, m-5-3*k,
      k], z, 5*d)), [x, y, z])};
      Systemez5d := Systemez5d union Coeffs_z5d[5, m-5-3*k, k];
  od:
  Systeme_z5d := Systemez5d:
  Variables_z5d := indets(Systeme_z5d):
  Solutions_z5d := solve(Systeme_z5d, Variables_z5d):
  assign(Solutions_z5d):

  for l from 0 to m-0*3 do nops(expand(AA||l)) od;
  for l from 0 to m-1*3 do nops(expand(BB||l)) od;
  for l from 0 to m-2*3 do nops(expand(CC||l)) od;
  for l from 0 to m-3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$\mathbf{1}$$

$$l := 3$$

$$\mathbf{1}$$

$$l := 4$$

$$\mathbf{1}$$

$$l := 5$$

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$l := 0$$

$$l := 0 \tag{7}$$

```
>
```